sg13g2_io_typ_1p2V_3p3V_25C Library

Cell Groups
INOUTx
INPUT
SG13G2_CORNER
SG13G2_FILLER10000
SG13G2_FILLER1000
SG13G2_FILLER2000
SG13G2_FILLER200
SG13G2_FILLER4000
SG13G2_FILLER400
SG13G2_IOPADANALOG
SG13G2_IOPADIOVDD
SG13G2_IOPADIOVSS
SG13G2_IOPADVDD
SG13G2_IOPADVSS
TRI_OUTx

INOUTx

sg13g2_io_typ_1p2V_3p3V_25C Cell Library: Process sg13g2_io_typ_1p2V_3p3V_25C, Voltage 1.20, Temp 25.00

Truth Table

INPUT			OUTPUT	
c2p	c2p_en	pad	pad	p2c
-	0	0	-	0
-	0	1	-	1
0	1	-	0	0
1	1	-	1	1

Footprint

Cell Name	Area
sg13g2_IOPadInOut16mA	14400.00000
sg13g2_IOPadInOut30mA	14400.00000
sg13g2_IOPadInOut4mA	14400.00000
sg13g2_IOPadOut16mA	14400.00000
sg13g2_IOPadOut30mA	14400.00000
sg13g2_IOPadOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	c2p	c2p_en	pad	p2c	pad
sg13g2_IOPadInOut16mA	0.02762	0.02589	0.29650	0.24000	20.00000
sg13g2_IOPadInOut30mA	0.02762	0.02589	0.38416	0.24000	40.00000
sg13g2_IOPadInOut4mA	0.02761	0.02589	0.22079	0.24000	12.00000
sg13g2_IOPadOut16mA	0.03902	0.00000	0.00000	0.00000	20.00000
sg13g2_IOPadOut30mA	0.03902	0.00000	0.00000	0.00000	40.00000
sg13g2_IOPadOut4mA	0.03902	0.00000	0.00000	0.00000	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadInOut16mA	47.17650	1103.23314	2043.55000
sg13g2_IOPadInOut30mA	165.33000	1248.19983	2043.32000
sg13g2_IOPadInOut4mA	80.75090	1039.52889	2043.34000
sg13g2_IOPadOut16mA	265.84800	447.63417	606.36100
sg13g2_IOPadOut30mA	411.75100	588.61867	893.54300
sg13g2_IOPadOut4mA	181.40700	359.67083	607.10400

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	**pad->p2c (RR)**	0.12000	0.02400	**0.22150**	0.80000	0.14400	**0.40366**	5.00000	0.24000	**0.68829**
sg13g2_IOPadInOut30mA	**pad->p2c (RR)**	0.12000	0.02400	**0.22118**	0.80000	0.14400	**0.40395**	5.00000	0.24000	**0.68914**
sg13g2_IOPadInOut4mA	**pad->p2c (RR)**	0.12000	0.02400	**0.22147**	0.80000	0.14400	**0.40365**	5.00000	0.24000	**0.68614**

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	**pad->p2c (FF)**	0.12000	0.02400	**0.21858**	0.80000	0.14400	**0.39677**	5.00000	0.24000	**0.71273**
sg13g2_IOPadInOut30mA	**pad->p2c (FF)**	0.12000	0.02400	**0.21870**	0.80000	0.14400	**0.39684**	5.00000	0.24000	**0.71289**
sg13g2_IOPadInOut4mA	**pad->p2c (FF)**	0.12000	0.02400	**0.21848**	0.80000	0.14400	**0.39644**	5.00000	0.24000	**0.71240**

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	**c2p->pad (RR)**	0.02000	1.00000	**1.55441**	0.33000	9.00000	**2.59085**	2.50000	20.00000	**3.89125**
	c2p_en->pad (FR)	0.02000	1.00000	**1.13779**	0.33000	9.00000	**1.21240**	2.50000	20.00000	**1.39959**
	c2p_en->pad (RR)	0.02000	1.00000	**1.55248**	0.33000	9.00000	**2.63715**	2.50000	20.00000	**4.00256**
sg13g2_IOPadInOut30mA	**c2p->pad (RR)**	0.02000	2.00000	**1.90123**	0.33000	18.00000	**3.15363**	2.50000	40.00000	**4.55693**
	c2p_en->pad (FR)	0.02000	2.00000	**1.33708**	0.33000	18.00000	**1.40555**	2.50000	40.00000	**1.59174**
	c2p_en->pad (RR)	0.02000	2.00000	**1.89540**	0.33000	18.00000	**3.20853**	2.50000	40.00000	**4.68587**
sg13g2_IOPadInOut4mA	**c2p->pad (RR)**	0.02000	1.00000	**1.57929**	0.33000	5.00000	**3.37312**	2.50000	12.00000	**6.50322**
	c2p_en->pad (FR)	0.02000	1.00000	**0.94889**	0.33000	5.00000	**1.02251**	2.50000	12.00000	**1.20089**
	c2p_en->pad (RR)	0.02000	1.00000	**1.58541**	0.33000	5.00000	**3.44452**	2.50000	12.00000	**6.69422**
sg13g2_IOPadOut16mA	**c2p->pad (RR)**	0.02000	1.00000	**1.48089**	0.33000	9.00000	**2.59607**	2.50000	20.00000	**4.84055**
sg13g2_IOPadOut30mA	**c2p->pad (RR)**	0.02000	2.00000	**1.82969**	0.33000	18.00000	**3.15987**	2.50000	40.00000	**5.50959**
sg13g2_IOPadOut4mA	**c2p->pad (RR)**	0.02000	1.00000	**1.49611**	0.33000	5.00000	**3.36624**	2.50000	12.00000	**7.44489**

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (FF)	0.02000	1.00000	**1.39190**	0.33000	9.00000	**2.32098**	2.50000	20.00000	**3.76195**
	c2p_en->pad (FF)	0.02000	1.00000	**1.35429**	0.33000	9.00000	**1.43264**	2.50000	20.00000	**1.69699**
	c2p_en->pad (RF)	0.02000	1.00000	**1.23670**	0.33000	9.00000	**2.26967**	2.50000	20.00000	**3.61070**
sg13g2_IOPadInOut30mA	c2p->pad (FF)	0.02000	2.00000	**1.75014**	0.33000	18.00000	**2.79256**	2.50000	40.00000	**4.27293**
	c2p_en->pad (FF)	0.02000	2.00000	**1.93939**	0.33000	18.00000	**2.02309**	2.50000	40.00000	**2.28201**
	c2p_en->pad (RF)	0.02000	2.00000	**1.40629**	0.33000	18.00000	**2.58385**	2.50000	40.00000	**4.00771**
sg13g2_IOPadInOut4mA	c2p->pad (FF)	0.02000	1.00000	**1.47206**	0.33000	5.00000	**3.25733**	2.50000	12.00000	**6.54271**
	c2p_en->pad (FF)	0.02000	1.00000	**0.87259**	0.33000	5.00000	**0.95690**	2.50000	12.00000	**1.22438**
	c2p_en->pad (RF)	0.02000	1.00000	**1.49390**	0.33000	5.00000	**3.40091**	2.50000	12.00000	**6.71586**
sg13g2_IOPadOut16mA	c2p->pad (FF)	0.02000	1.00000	**1.29845**	0.33000	9.00000	**2.18744**	2.50000	20.00000	**3.45651**
sg13g2_IOPadOut30mA	c2p->pad (FF)	0.02000	2.00000	**1.66081**	0.33000	18.00000	**2.66725**	2.50000	40.00000	**3.97305**
sg13g2_IOPadOut4mA	c2p->pad (FF)	0.02000	1.00000	**1.35220**	0.33000	5.00000	**3.10361**	2.50000	12.00000	**6.21594**

Power Information
Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	**pad**	0.12000	0.02400	**-1.27814**	0.80000	0.14400	**-1.16405**	5.00000	0.24000	**-0.33642**
	pad	0.12000	0.02400	**0.02797**	0.80000	0.14400	**0.02933**	5.00000	0.24000	**0.03171**
sg13g2_IOPadInOut30mA	**pad**	0.12000	0.02400	**-2.00002**	0.80000	0.14400	**-1.88070**	5.00000	0.24000	**-1.05253**
	pad	0.12000	0.02400	**0.02808**	0.80000	0.14400	**0.02975**	5.00000	0.24000	**0.03211**
sg13g2_IOPadInOut4mA	**pad**	0.12000	0.02400	**-0.69496**	0.80000	0.14400	**-0.58285**	5.00000	0.24000	**0.24728**
	pad	0.12000	0.02400	**0.02777**	0.80000	0.14400	**0.02915**	5.00000	0.24000	**0.03162**

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	**pad**	0.12000	0.02400	**1.75063**	0.80000	0.14400	**1.85788**	5.00000	0.24000	**2.69362**
	pad	0.12000	0.02400	**0.07078**	0.80000	0.14400	**0.07927**	5.00000	0.24000	**0.11232**
sg13g2_IOPadInOut30mA	**pad**	0.12000	0.02400	**2.46845**	0.80000	0.14400	**2.57036**	5.00000	0.24000	**3.40958**
	pad	0.12000	0.02400	**0.06850**	0.80000	0.14400	**0.07683**	5.00000	0.24000	**0.10996**
sg13g2_IOPadInOut4mA	**pad**	0.12000	0.02400	**1.16627**	0.80000	0.14400	**1.27890**	5.00000	0.24000	**2.11079**
	pad	0.12000	0.02400	**0.06887**	0.80000	0.14400	**0.07733**	5.00000	0.24000	**0.11099**

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	**6.54690**	0.33000	9.00000	**6.48434**	2.50000	20.00000	**6.60733**
	c2p	0.02000	1.00000	**1.49731**	0.33000	9.00000	**12.90800**	2.50000	20.00000	**14.70460**
	c2p_en	0.02000	1.00000	**6.25504**	0.33000	9.00000	**8.74216**	2.50000	20.00000	**11.66930**
	c2p_en	0.02000	1.00000	**1.46754**	0.33000	9.00000	**12.84770**	2.50000	20.00000	**14.69630**
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	**10.10630**	0.33000	18.00000	**9.52861**	2.50000	40.00000	**8.62653**
	c2p	0.02000	2.00000	**2.92518**	0.33000	18.00000	**21.18490**	2.50000	40.00000	**1.40112**
	c2p_en	0.02000	2.00000	**9.61326**	0.33000	18.00000	**14.48540**	2.50000	40.00000	**20.71970**
	c2p_en	0.02000	2.00000	**2.89629**	0.33000	18.00000	**21.18340**	2.50000	40.00000	**1.27998**
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	**3.80926**	0.33000	5.00000	**4.15331**	2.50000	12.00000	**5.41203**
	c2p	0.02000	1.00000	**1.50493**	0.33000	5.00000	**7.18811**	2.50000	12.00000	**17.32470**
	c2p_en	0.02000	1.00000	**3.58012**	0.33000	5.00000	**4.79200**	2.50000	12.00000	**6.92417**
	c2p_en	0.02000	1.00000	**1.47492**	0.33000	5.00000	**7.16155**	2.50000	12.00000	**17.23550**
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	**6.15045**	0.33000	9.00000	**6.01115**	2.50000	20.00000	**5.85909**
	c2p	0.02000	1.00000	**-0.01698**	0.33000	9.00000	**-0.01451**	2.50000	20.00000	**0.05008**
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	**9.80602**	0.33000	18.00000	**9.34405**	2.50000	40.00000	**9.10140**
	c2p	0.02000	2.00000	**-0.01698**	0.33000	18.00000	**-0.01451**	2.50000	40.00000	**0.05006**
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	**3.22417**	0.33000	5.00000	**3.11329**	2.50000	12.00000	**3.08326**
	c2p	0.02000	1.00000	**-0.01698**	0.33000	5.00000	**-0.01451**	2.50000	12.00000	**0.04977**

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	**14.17070**	0.33000	9.00000	**9.57559**	2.50000	20.00000	**8.33041**
	c2p	0.02000	1.00000	**0.19384**	0.33000	9.00000	**0.20684**	2.50000	20.00000	**0.27988**
	c2p_en	0.02000	1.00000	**3.66816**	0.33000	9.00000	**3.92688**	2.50000	20.00000	**4.33549**
	c2p_en	0.02000	1.00000	**0.10804**	0.33000	9.00000	**0.12318**	2.50000	20.00000	**0.17272**
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	**51.47480**	0.33000	18.00000	**31.52110**	2.50000	40.00000	**23.83460**
	c2p	0.02000	2.00000	**0.19620**	0.33000	18.00000	**0.20752**	2.50000	40.00000	**0.27266**
	c2p_en	0.02000	2.00000	**5.49394**	0.33000	18.00000	**5.77125**	2.50000	40.00000	**6.19919**
	c2p_en	0.02000	2.00000	**0.10789**	0.33000	18.00000	**0.12377**	2.50000	40.00000	**0.16576**
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	**2.93847**	0.33000	5.00000	**3.52597**	2.50000	12.00000	**4.56238**
	c2p	0.02000	1.00000	**0.20006**	0.33000	5.00000	**0.22312**	2.50000	12.00000	**0.31443**
	c2p_en	0.02000	1.00000	**2.25901**	0.33000	5.00000	**2.84683**	2.50000	12.00000	**3.88665**
	c2p_en	0.02000	1.00000	**0.11500**	0.33000	5.00000	**0.13941**	2.50000	12.00000	**0.20688**
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	**15.33040**	0.33000	9.00000	**9.45038**	2.50000	20.00000	**7.68065**
	c2p	0.02000	1.00000	**0.06107**	0.33000	9.00000	**0.06644**	2.50000	20.00000	**0.13159**
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	**54.41300**	0.33000	18.00000	**32.39940**	2.50000	40.00000	**23.95270**
	c2p	0.02000	2.00000	**0.06105**	0.33000	18.00000	**0.06644**	2.50000	40.00000	**0.13130**
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	**2.52595**	0.33000	5.00000	**2.43756**	2.50000	12.00000	**2.40009**
	c2p	0.02000	1.00000	**0.06107**	0.33000	5.00000	**0.06646**	2.50000	12.00000	**0.13377**

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	**-0.00007**	0.33000	**-0.00007**	2.50000	**-0.00007**
	0.02000	**-0.01828**	0.33000	**-0.01924**	2.50000	**-0.01939**
sg13g2_IOPadInOut30mA	0.02000	**-0.00008**	0.33000	**-0.00007**	2.50000	**-0.00007**
	0.02000	**-0.01828**	0.33000	**-0.01924**	2.50000	**-0.01939**
sg13g2_IOPadInOut4mA	0.02000	**-0.00008**	0.33000	**-0.00007**	2.50000	**-0.00007**
	0.02000	**-0.01828**	0.33000	**-0.01925**	2.50000	**-0.01939**

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	**0.00007**	0.33000	**0.00007**	2.50000	**0.00007**
	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**
sg13g2_IOPadInOut30mA	0.02000	**0.00008**	0.33000	**0.00007**	2.50000	**0.00007**
	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**
sg13g2_IOPadInOut4mA	0.02000	**0.00008**	0.33000	**0.00007**	2.50000	**0.00007**
	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	**(!c2p_en * pad * p2c)**	0.02000	**-0.00007**	0.33000	**-0.00006**	2.50000	**-0.00007**
	(!c2p_en * pad * p2c)	0.02000	**-0.01828**	0.33000	**-0.01924**	2.50000	**-0.01939**
	(!c2p_en * !pad * !p2c)	0.02000	**-0.00007**	0.33000	**-0.00007**	2.50000	**-0.00007**
	(!c2p_en * !pad * !p2c)	0.02000	**-0.01828**	0.33000	**-0.01924**	2.50000	**-0.01939**
sg13g2_IOPadInOut30mA	**(!c2p_en * pad * p2c)**	0.02000	**-0.00007**	0.33000	**-0.00006**	2.50000	**-0.00007**
	(!c2p_en * pad * p2c)	0.02000	**-0.01828**	0.33000	**-0.01924**	2.50000	**-0.01939**
	(!c2p_en * !pad * !p2c)	0.02000	**-0.00008**	0.33000	**-0.00007**	2.50000	**-0.00007**
	(!c2p_en * !pad * !p2c)	0.02000	**-0.01828**	0.33000	**-0.01924**	2.50000	**-0.01939**
sg13g2_IOPadInOut4mA	**(!c2p_en * pad * p2c)**	0.02000	**-0.00007**	0.33000	**-0.00006**	2.50000	**-0.00007**
	(!c2p_en * pad * p2c)	0.02000	**-0.01828**	0.33000	**-0.01925**	2.50000	**-0.01939**
	(!c2p_en * !pad * !p2c)	0.02000	**-0.00008**	0.33000	**-0.00007**	2.50000	**-0.00007**
	(!c2p_en * !pad * !p2c)	0.02000	**-0.01828**	0.33000	**-0.01925**	2.50000	**-0.01939**

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	**(!c2p_en * pad * p2c)**	0.02000	**0.00007**	0.33000	**0.00006**	2.50000	**0.00007**
	(!c2p_en * pad * p2c)	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**
	(!c2p_en * !pad * !p2c)	0.02000	**0.00007**	0.33000	**0.00007**	2.50000	**0.00007**
	(!c2p_en * !pad * !p2c)	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**
sg13g2_IOPadInOut30mA	**(!c2p_en * pad * p2c)**	0.02000	**0.00007**	0.33000	**0.00006**	2.50000	**0.00007**
	(!c2p_en * pad * p2c)	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**
	(!c2p_en * !pad * !p2c)	0.02000	**0.00008**	0.33000	**0.00007**	2.50000	**0.00007**
	(!c2p_en * !pad * !p2c)	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**
sg13g2_IOPadInOut4mA	**(!c2p_en * pad * p2c)**	0.02000	**0.00007**	0.33000	**0.00006**	2.50000	**0.00007**
	(!c2p_en * pad * p2c)	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**
	(!c2p_en * !pad * !p2c)	0.02000	**0.00008**	0.33000	**0.00007**	2.50000	**0.00007**
	(!c2p_en * !pad * !p2c)	0.02000	**0.02424**	0.33000	**0.02411**	2.50000	**0.02394**

Passive power(pJ) for c2p_en rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	**1.66710**	0.33000	**1.66704**	2.50000	**1.67489**
	0.02000	**0.03254**	0.33000	**0.03264**	2.50000	**0.06907**
sg13g2_IOPadInOut30mA	0.02000	**2.61137**	0.33000	**2.61232**	2.50000	**2.62178**
	0.02000	**0.03257**	0.33000	**0.03264**	2.50000	**0.06899**
sg13g2_IOPadInOut4mA	0.02000	**0.86396**	0.33000	**0.86411**	2.50000	**0.86631**
	0.02000	**0.03254**	0.33000	**0.03265**	2.50000	**0.06909**

Passive power(pJ) for c2p_en falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	**0.35585**	0.33000	**0.35491**	2.50000	**0.35254**
	0.02000	**0.08502**	0.33000	**0.08604**	2.50000	**0.12225**
sg13g2_IOPadInOut30mA	0.02000	**0.34683**	0.33000	**0.34612**	2.50000	**0.34437**
	0.02000	**0.08501**	0.33000	**0.08605**	2.50000	**0.12239**
sg13g2_IOPadInOut4mA	0.02000	**0.37919**	0.33000	**0.37836**	2.50000	**0.37694**
	0.02000	**0.08502**	0.33000	**0.08604**	2.50000	**0.12240**

Passive power(pJ) for pad rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	**6.54690**	0.33000	**6.48434**	2.50000	**6.60733**
	0.02000	**1.49731**	0.33000	**12.90800**	2.50000	**14.70460**
	0.02000	**6.25504**	0.33000	**8.74216**	2.50000	**11.66930**
	0.02000	**1.46754**	0.33000	**12.84770**	2.50000	**14.69630**
sg13g2_IOPadInOut30mA	0.02000	**10.10630**	0.33000	**9.52861**	2.50000	**8.62653**
	0.02000	**2.92518**	0.33000	**21.18490**	2.50000	**1.40112**
	0.02000	**9.61326**	0.33000	**14.48540**	2.50000	**20.71970**
	0.02000	**2.89629**	0.33000	**21.18340**	2.50000	**1.27998**
sg13g2_IOPadInOut4mA	0.02000	**3.80926**	0.33000	**4.15331**	2.50000	**5.41203**
	0.02000	**1.50493**	0.33000	**7.18811**	2.50000	**17.32470**
	0.02000	**3.58012**	0.33000	**4.79200**	2.50000	**6.92417**
	0.02000	**1.47492**	0.33000	**7.16155**	2.50000	**17.23550**
sg13g2_IOPadOut16mA	0.02000	**6.15045**	0.33000	**6.01115**	2.50000	**5.85909**
	0.02000	**-0.01698**	0.33000	**-0.01451**	2.50000	**0.05008**
sg13g2_IOPadOut30mA	0.02000	**9.80602**	0.33000	**9.34405**	2.50000	**9.10140**
	0.02000	**-0.01698**	0.33000	**-0.01451**	2.50000	**0.05006**
sg13g2_IOPadOut4mA	0.02000	**3.22417**	0.33000	**3.11329**	2.50000	**3.08326**
	0.02000	**-0.01698**	0.33000	**-0.01451**	2.50000	**0.04977**

Passive power(pJ) for pad falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	**14.17070**	0.33000	**9.57559**	2.50000	**8.33041**
	0.02000	**0.19384**	0.33000	**0.20684**	2.50000	**0.27988**
	0.02000	**3.66816**	0.33000	**3.92688**	2.50000	**4.33549**
	0.02000	**0.10804**	0.33000	**0.12318**	2.50000	**0.17272**
sg13g2_IOPadInOut30mA	0.02000	**51.47480**	0.33000	**31.52110**	2.50000	**23.83460**
	0.02000	**0.19620**	0.33000	**0.20752**	2.50000	**0.27266**
	0.02000	**5.49394**	0.33000	**5.77125**	2.50000	**6.19919**
	0.02000	**0.10789**	0.33000	**0.12377**	2.50000	**0.16576**
sg13g2_IOPadInOut4mA	0.02000	**2.93847**	0.33000	**3.52597**	2.50000	**4.56238**
	0.02000	**0.20006**	0.33000	**0.22312**	2.50000	**0.31443**
	0.02000	**2.25901**	0.33000	**2.84683**	2.50000	**3.88665**
	0.02000	**0.11500**	0.33000	**0.13941**	2.50000	**0.20688**
sg13g2_IOPadOut16mA	0.02000	**15.33040**	0.33000	**9.45038**	2.50000	**7.68065**
	0.02000	**0.06107**	0.33000	**0.06644**	2.50000	**0.13159**
sg13g2_IOPadOut30mA	0.02000	**54.41300**	0.33000	**32.39940**	2.50000	**23.95270**
	0.02000	**0.06105**	0.33000	**0.06644**	2.50000	**0.13130**
sg13g2_IOPadOut4mA	0.02000	**2.52595**	0.33000	**2.43756**	2.50000	**2.40009**
	0.02000	**0.06107**	0.33000	**0.06646**	2.50000	**0.13377**

INPUT

sg13g2_io_typ_1p2V_3p3V_25C Cell Library: Process sg13g2_io_typ_1p2V_3p3V_25C, Voltage 1.20, Temp 25.00

Truth Table

INPUT	OUTPUT
pad	p2c
0	0
1	1

Footprint

Cell Name	Area
sg13g2_IOPadIn	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	pad	p2c
sg13g2_IOPadIn	0.23445	0.24000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadIn	108.10100	183.82567	308.12900

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	**pad->p2c (RR)**	0.12000	0.02400	**0.22121**	0.80000	0.14400	**0.40324**	5.00000	0.24000	**0.68563**

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	**pad->p2c (FF)**	0.12000	0.02400	**0.21837**	0.80000	0.14400	**0.39638**	5.00000	0.24000	**0.71182**

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	**-0.60889**	0.80000	0.14400	**-0.49587**	5.00000	0.24000	**0.32960**
	pad	0.12000	0.02400	**0.02849**	0.80000	0.14400	**0.02997**	5.00000	0.24000	**0.03229**

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	**1.08038**	0.80000	0.14400	**1.19245**	5.00000	0.24000	**2.02702**
	pad	0.12000	0.02400	**0.06800**	0.80000	0.14400	**0.07651**	5.00000	0.24000	**0.10991**

SG13G2_CORNER

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Corner	32400.00000

Pin Capacitance Information

SG13G2_FILLER10000

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler10000	18000.00000

Pin Capacitance Information

SG13G2_FILLER1000

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler1000	900.00000

Pin Capacitance Information

SG13G2_FILLER2000

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler2000	1800.00000

Pin Capacitance Information

SG13G2_FILLER200

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler200	180.00000

Pin Capacitance Information

SG13G2_FILLER4000

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler4000	3600.00000

Pin Capacitance Information

SG13G2_FILLER400

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler400	360.00000

Pin Capacitance Information

SG13G2_IOPADANALOG

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Truth Table

INPUT		OUTPUT	
pad	padres	pad	padres
x	x	-	-

Footprint

Cell Name	Area
sg13g2_IOPadAnalog	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	pad	padres	pad	padres
sg13g2_IOPadAnalog	0.76624	0.60557	30000.00000	30000.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadAnalog	0.00000	517.45000	1034.90000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	**pad->pad (R)**	10.00000	500.00000	**1000.00000**	200.00000	30000.00000	**1000.00000**	200.00000	30000.00000	**1000.00000**

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	**pad->pad (F)**	10.00000	500.00000	**1000.00000**	200.00000	30000.00000	**1000.00000**	200.00000	30000.00000	**1000.00000**

SG13G2_IOPADIOVDD

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADIOVSS

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVss	14400.00000

Pin Capacitance Information

SG13G2_IOPADVDD

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADVSS

sg13g2_io_typ_1p2V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p2V_3p3V_25C,
Voltage 1.20, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadVss	14400.00000

Pin Capacitance Information

TRI_OUTx

sg13g2_io_typ_1p2V_3p3V_25C Cell Library: Process sg13g2_io_typ_1p2V_3p3V_25C, Voltage 1.20, Temp 25.00

Truth Table

INPUT		OUTPUT
c2p	c2p_en	pad
-	0	HiZ
0	1	0
1	1	1

Footprint

Cell Name	Area
sg13g2_IOPadTriOut16mA	14400.00000
sg13g2_IOPadTriOut30mA	14400.00000
sg13g2_IOPadTriOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	c2p	c2p_en	pad
sg13g2_IOPadTriOut16mA	0.02724	0.02633	20.00000
sg13g2_IOPadTriOut30mA	0.02724	0.02633	40.00000
sg13g2_IOPadTriOut4mA	0.02724	0.02634	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadTriOut16mA	140.83000	945.36450	1754.75000
sg13g2_IOPadTriOut30mA	266.32900	1046.41070	1755.85000
sg13g2_IOPadTriOut4mA	51.12180	865.76980	1754.69000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (RR)	0.02000	1.00000	**1.55305**	0.33000	9.00000	**2.59220**	2.50000	20.00000	**3.89266**
	c2p_en->pad (FR)	0.02000	1.00000	**1.14782**	0.33000	9.00000	**1.22036**	2.50000	20.00000	**1.40850**
	c2p_en->pad (RR)	0.02000	1.00000	**1.55418**	0.33000	9.00000	**2.64307**	2.50000	20.00000	**4.01099**
sg13g2_IOPadTriOut30mA	c2p->pad (RR)	0.02000	2.00000	**1.90147**	0.33000	18.00000	**3.15575**	2.50000	40.00000	**4.56130**
	c2p_en->pad (FR)	0.02000	2.00000	**1.34400**	0.33000	18.00000	**1.41725**	2.50000	40.00000	**1.60086**
	c2p_en->pad (RR)	0.02000	2.00000	**1.89876**	0.33000	18.00000	**3.21663**	2.50000	40.00000	**4.69588**
sg13g2_IOPadTriOut4mA	c2p->pad (RR)	0.02000	1.00000	**1.56957**	0.33000	5.00000	**3.36308**	2.50000	12.00000	**6.49233**
	c2p_en->pad (FR)	0.02000	1.00000	**0.95881**	0.33000	5.00000	**1.03146**	2.50000	12.00000	**1.21649**
	c2p_en->pad (RR)	0.02000	1.00000	**1.57984**	0.33000	5.00000	**3.44586**	2.50000	12.00000	**6.70248**

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (FF)	0.02000	1.00000	**1.39604**	0.33000	9.00000	**2.32622**	2.50000	20.00000	**3.76773**
	c2p_en->pad (FF)	0.02000	1.00000	**1.36649**	0.33000	9.00000	**1.44731**	2.50000	20.00000	**1.70684**
	c2p_en->pad (RF)	0.02000	1.00000	**1.24069**	0.33000	9.00000	**2.27806**	2.50000	20.00000	**3.62113**
sg13g2_IOPadTriOut30mA	c2p->pad (FF)	0.02000	2.00000	**1.75700**	0.33000	18.00000	**2.80173**	2.50000	40.00000	**4.27930**
	c2p_en->pad (FF)	0.02000	2.00000	**1.91682**	0.33000	18.00000	**2.03459**	2.50000	40.00000	**2.27137**
	c2p_en->pad (RF)	0.02000	2.00000	**1.41148**	0.33000	18.00000	**2.59244**	2.50000	40.00000	**4.02396**
sg13g2_IOPadTriOut4mA	c2p->pad (FF)	0.02000	1.00000	**1.46603**	0.33000	5.00000	**3.25093**	2.50000	12.00000	**6.53598**
	c2p_en->pad (FF)	0.02000	1.00000	**0.88100**	0.33000	5.00000	**0.96469**	2.50000	12.00000	**1.23308**
	c2p_en->pad (RF)	0.02000	1.00000	**1.49122**	0.33000	5.00000	**3.39980**	2.50000	12.00000	**6.71939**

Power Information

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	**c2p**	0.02000	1.00000	**6.10011**	0.33000	9.00000	**5.96930**	2.50000	20.00000	**5.64409**
	c2p	0.02000	1.00000	**0.05199**	0.33000	9.00000	**0.05059**	2.50000	20.00000	**0.10987**
	c2p_en	0.02000	1.00000	**5.77818**	0.33000	9.00000	**8.24812**	2.50000	20.00000	**11.36840**
	c2p_en	0.02000	1.00000	**0.02213**	0.33000	9.00000	**0.02351**	2.50000	20.00000	**0.09601**
sg13g2_IOPadTriOut30mA	**c2p**	0.02000	2.00000	**9.73879**	0.33000	18.00000	**9.28333**	2.50000	40.00000	**8.89053**
	c2p	0.02000	2.00000	**0.05200**	0.33000	18.00000	**0.05059**	2.50000	40.00000	**0.10983**
	c2p_en	0.02000	2.00000	**9.18239**	0.33000	18.00000	**14.31020**	2.50000	40.00000	**20.90680**
	c2p_en	0.02000	2.00000	**0.02213**	0.33000	18.00000	**0.02351**	2.50000	40.00000	**0.09601**
sg13g2_IOPadTriOut4mA	**c2p**	0.02000	1.00000	**3.21222**	0.33000	5.00000	**3.08969**	2.50000	12.00000	**2.89552**
	c2p	0.02000	1.00000	**0.05202**	0.33000	5.00000	**0.05063**	2.50000	12.00000	**0.10987**
	c2p_en	0.02000	1.00000	**2.98525**	0.33000	5.00000	**3.79246**	2.50000	12.00000	**5.20413**
	c2p_en	0.02000	1.00000	**0.02216**	0.33000	5.00000	**0.02354**	2.50000	12.00000	**0.09458**

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	**c2p**	0.02000	1.00000	**13.76330**	0.33000	9.00000	**8.92355**	2.50000	20.00000	**7.27147**
	c2p	0.02000	1.00000	**0.11999**	0.33000	9.00000	**0.11924**	2.50000	20.00000	**0.17975**
	c2p_en	0.02000	1.00000	**3.27268**	0.33000	9.00000	**3.27155**	2.50000	20.00000	**3.27056**
	c2p_en	0.02000	1.00000	**0.03502**	0.33000	9.00000	**0.03525**	2.50000	20.00000	**0.07255**
sg13g2_IOPadTriOut30mA	**c2p**	0.02000	2.00000	**50.89800**	0.33000	18.00000	**30.82310**	2.50000	40.00000	**22.66180**
	c2p	0.02000	2.00000	**0.12000**	0.33000	18.00000	**0.11926**	2.50000	40.00000	**0.17987**
	c2p_en	0.02000	2.00000	**5.08903**	0.33000	18.00000	**5.09225**	2.50000	40.00000	**5.09356**
	c2p_en	0.02000	2.00000	**0.03504**	0.33000	18.00000	**0.03526**	2.50000	40.00000	**0.07246**
sg13g2_IOPadTriOut4mA	**c2p**	0.02000	1.00000	**2.45234**	0.33000	5.00000	**2.45370**	2.50000	12.00000	**2.45773**
	c2p	0.02000	1.00000	**0.12014**	0.33000	5.00000	**0.11940**	2.50000	12.00000	**0.17987**
	c2p_en	0.02000	1.00000	**1.75684**	0.33000	5.00000	**1.75522**	2.50000	12.00000	**1.75430**
	c2p_en	0.02000	1.00000	**0.03516**	0.33000	5.00000	**0.03540**	2.50000	12.00000	**0.07242**

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	0.02000	**-0.01815**	0.33000	**-0.01926**	2.50000	**-0.01950**
sg13g2_IOPadTriOut30mA	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	0.02000	**-0.01815**	0.33000	**-0.01926**	2.50000	**-0.01950**
sg13g2_IOPadTriOut4mA	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	0.02000	**-0.01815**	0.33000	**-0.01926**	2.50000	**-0.01950**

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	0.02000	**0.02323**	0.33000	**0.02303**	2.50000	**0.02279**
sg13g2_IOPadTriOut30mA	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	0.02000	**0.02323**	0.33000	**0.02303**	2.50000	**0.02279**
sg13g2_IOPadTriOut4mA	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	0.02000	**0.02323**	0.33000	**0.02303**	2.50000	**0.02247**

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	**!c2p_en**	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	!c2p_en	0.02000	**-0.01815**	0.33000	**-0.01926**	2.50000	**-0.01950**
sg13g2_IOPadTriOut30mA	**!c2p_en**	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	!c2p_en	0.02000	**-0.01815**	0.33000	**-0.01926**	2.50000	**-0.01950**
sg13g2_IOPadTriOut4mA	**!c2p_en**	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	!c2p_en	0.02000	**-0.01815**	0.33000	**-0.01926**	2.50000	**-0.01950**

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	!c2p_en	0.02000	**0.02323**	0.33000	**0.02303**	2.50000	**0.02279**
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	!c2p_en	0.02000	**0.02323**	0.33000	**0.02303**	2.50000	**0.02279**
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	**0.00000**	0.33000	**0.00000**	2.50000	**0.00000**
	!c2p_en	0.02000	**0.02323**	0.33000	**0.02303**	2.50000	**0.02247**